Ch 1: Chemical Reactions and Equations

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SECTION A - 2 MARKS

Q1. What is a chemical equation? What information does a balanced chemical equation convey?

Q2. What is meant by a balanced chemical equation? Why should chemical equations be balanced? (PYQ 2022)

Q3. Name the type of chemical reaction that takes place when quicklime is added to water. Write the chemical equation for this reaction. (PYQ 2021)

Q4. Give the chemical name of the reactants and products of the following equation: HNO3 + Ca(OH)2 -> Ca(NO3)2 + H2O. (PYQ 2021)

Q5. Assertion (A): Burning of natural gas is an exothermic process. Reason (R): Methane gas combines with oxygen to produce carbon dioxide and water. Choose the correct option and justify. (PYQ 2023)

Q6. What happens when potassium iodide solution is mixed with lead nitrate solution? Write the balanced chemical equation. (PYQ 2023)

Q7. What is a combination reaction? Give one example with a balanced chemical equation.

Q8. What is a decomposition reaction? How is it opposite to a combination reaction? Give one example.

Q9. What is a displacement reaction? Give one example with a balanced chemical equation.

Q10. What is a double displacement reaction? How is it different from a displacement reaction? (PYQ 2019)

Q11. What is a precipitation reaction? Give one example with a balanced equation.

Q12. What is an exothermic reaction? Give two examples of exothermic reactions. (PYQ 2018)

Q13. What is an endothermic reaction? Give two examples. How is it different from an exothermic reaction?

Q14. What is oxidation? What is reduction? Define with one example each showing both occurring simultaneously.

Q15. Which substance is oxidised and which is reduced in the following reaction? CuO + H2 -> Cu + H2O. (PYQ 2020)

Q16. What is a redox reaction? Identify the oxidising agent and reducing agent in: MnO2 + 4HCl -> MnCl2 + 2H2O + Cl2. (PYQ 2019)

Q17. Why does the colour of copper sulphate solution change when an iron nail is placed in it? Write the chemical equation. (PYQ 2018)

Q18. What is corrosion? Write the chemical equation for the corrosion of iron. What are the conditions required for it?

Q19. What is rancidity? How can rancidity be prevented? Give two methods. (PYQ 2019)

Q20. Why are chips manufacturers usually flush bags of chips with nitrogen gas? (PYQ 2020)

Q21. Why is it necessary to balance a chemical equation? State the law it is based on. (PYQ 2022)

Q22. Identify the type of reaction: 2AgCl(s) -> 2Ag(s) + Cl2(g). State the condition required and one practical application. (PYQ 2024)

Q23. A pale green substance A is heated. A brown residue B forms along with two gases having the smell of burning sulphur. Identify A and B. Write the chemical equation. (PYQ HOTS)

Q24. What are state symbols in a chemical equation? Write the state symbols for solid, liquid, gas, and aqueous solution.

Q25. Why is magnesium stored in dry sand? What type of reaction does magnesium undergo when it burns in air?

Q26. Balance the following chemical equation: $Fe + H_2O \rightarrow Fe_3O_4 + H_2$. (PYQ 2022)

Q27. Balance the following chemical equation: $KClO_3 \rightarrow KCl + O_2$. Write the state symbols. (PYQ 2020)

Q28. Write the balanced chemical equation for the electrolytic decomposition of water. What type of reaction is it?

Q29. What is a photolytic decomposition reaction? Give one example with a balanced equation.

Q30. What is the difference between thermal decomposition and electrolytic decomposition? Give one example each.

Q31. Identify the oxidising agent in the following reaction: $2PbO + C \rightarrow 2Pb + CO_2$. Justify your answer. (PYQ 2019)

Q32. Why does silver chloride turn grey when kept in sunlight? Write the chemical equation for the reaction. (PYQ 2024)

SECTION B - 3 MARKS

Q1. Write the balanced chemical equations for the following reactions and identify the type of each reaction: (PYQ 2019)

Reaction	Balanced Equation	Type of Reaction
Magnesium burns in oxygen		
HCl added to Na_2CO_3		
Iron displaces copper from $CuSO_4$		

Q2. Write the balanced chemical equations for the following and identify the type of reaction: (PYQ 2018)

Reaction	Balanced Equation	Type of Reaction
Potassium bromide + Barium iodide		
Zinc + Silver nitrate solution		
Barium chloride + Aluminium sulphate		

Q3. Write the balanced chemical equations for the following reactions. Identify the type of each reaction: (PYQ 2022)

Reaction	Balanced Equation	Type
$AgNO_3(aq) + NaCl(aq)$		
$Pb(NO_3)_2(aq) + 2KI(aq)$		
$Na_2SO_4(aq) + BaCl_2(aq)$		

Q4. In each of the following reactions, identify which substance is oxidised and which is reduced. Also identify the oxidising agent and reducing agent: (PYQ 2020)

Reaction	Oxidised	Reduced	Oxidising Agent	Reducing Agent
$CuO + H_2 \rightarrow Cu + H_2O$				
$2PbO + C \rightarrow 2Pb + CO_2$				
$MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$				

Q5. Balance the following chemical equations and identify the type of each reaction: (PYQ 2019)

Unbalanced Equation	Balanced Equation	Type of Reaction
$Fe + Cl_2 \rightarrow FeCl_3$		
$Mg + N_2 \rightarrow Mg_3N_2$		
$H_2SO_4 + NaOH \rightarrow Na_2SO_4 + H_2O$		

Q6. Write the balanced equations and identify the type of reaction for each of the following: (PYQ 2023)

Reaction	Balanced Equation	Type
Quicklime + water		
Ferrous sulphate heated		
Lead nitrate solution + potassium iodide solution		

Q7. Write one balanced equation each for the following types of reactions: (PYQ 2018)

Type of Reaction	Example — Balanced Chemical Equation
Combination reaction	
Decomposition reaction (thermal)	
Displacement reaction	

Q8. Write one balanced equation each for the following types of reactions: (PYQ 2019)

Type of Reaction	Example — Balanced Chemical Equation
Double displacement reaction	
Exothermic reaction	
Endothermic reaction	

Q9. Identify the type of reaction in each of the following and give a reason: (PYQ 2022)

Reaction	Type of Reaction	Reason
$2H_2 + O_2 \rightarrow 2H_2O$		
$2AgCl \rightarrow 2Ag + Cl_2$ (sunlight)		
$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$		

Q10. State what happens in each of the following cases and write a balanced equation where applicable: (PYQ 2020)

Situation	Observation	Balanced Equation
Iron nail placed in copper sulphate solution		
Copper powder heated in china dish		
Silver article left in air for a few weeks		

Q11. Write balanced chemical equations for the following decomposition reactions: (PYQ 2019)

Substance Decomposed	Balanced Equation	Energy Required
Calcium carbonate (heated)		
Lead nitrate (heated)		
Silver bromide (exposed to light)		

Q12. Identify the type of reaction and write the balanced chemical equation for each: (PYQ 2024)

Reaction	Balanced Equation	Type
Slaked lime + $CO_2 \rightarrow$ calcium carbonate + water		
Copper + $2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$		
$BaCl_2(aq) + H_2SO_4(aq) \rightarrow BaSO_4(s) + 2HCl(aq)$		

Q13. Write the balanced chemical equation for the following reactions. Classify each as combination, decomposition, displacement, or double displacement: (PYQ 2020)

Reaction	Balanced Equation	Classification
Zinc + dilute sulphuric acid		
Magnesium ribbon + oxygen		
Sodium hydroxide + hydrochloric acid		

Q14. For the following reactions, state the type of reaction and identify the substance being oxidised and reduced: (PYQ 2019)

Reaction	Type	Substance Oxidised	Substance Reduced
$ZnO + C \rightarrow Zn + CO$			
$H_2S + Cl_2 \rightarrow 2HCl + S$			
$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$			

Q15. Write the balanced chemical equations for the following with state symbols. Identify the type:

Reaction	Balanced Equation (with State Symbols)	Type
Lead sulphide + oxygen -> lead oxide + sulphur dioxide		
Ferric oxide + hydrochloric acid		
Sodium sulphate + barium chloride solution		

Q16. What is a redox reaction? Write two examples of redox reactions showing both oxidation and reduction simultaneously. Identify the oxidising and reducing agents in each. (PYQ 2019)

Q17. What is corrosion? Explain with two examples. Write the chemical equation for the corrosion of iron. What are the conditions required? (PYQ 2020)

Q18. What is rancidity? Explain why fats and oils become rancid. State three methods to prevent rancidity. (PYQ 2019)

Q19. Write the balanced chemical equations for the following and state one observation for each: (PYQ 2022)

Reaction	Balanced Equation	Observation
Iron + steam -> iron oxide + hydrogen		
Silver chloride in sunlight		
Ferrous sulphate crystals on strong heating		

Q20. Balance the following equations and write the type of each reaction: (PYQ 2018)

Unbalanced Equation	Balanced Equation	Type
Ca + O2 -> CaO		
Na + H2O -> NaOH + H2		
AgNO3 + NaCl -> AgCl + NaNO3		

Q21. What happens when: (i) iron is kept in copper sulphate solution (ii) zinc is added to dilute hydrochloric acid (iii) silver nitrate is mixed with sodium chloride solution. Write equations for each. (PYQ 2020)

Q22. Write the essential conditions for the following reaction and name its type: 2AgCl -> 2Ag + Cl2. Also write one practical use of this reaction. (PYQ 2024)

Q23. Write balanced chemical equations for the following and identify the type of reaction in each: (PYQ 2023)

Reaction	Balanced Equation	Type
Nitrogen dioxide + water -> nitric acid + nitric oxide		
Copper + silver nitrate solution		
Barium hydroxide + dilute H2SO4		

Q24. A student adds a few drops of sulphuric acid to zinc granules in a test tube. State two observations and write the balanced chemical equation. Name the type of reaction and the gas evolved. (PYQ)

Q25. Identify the type of reaction and substances oxidised and reduced in each: (PYQ 2019)

Reaction	Type	Substance Oxidised	Substance Reduced
$4Na + O_2 \rightarrow 2Na_2O$			
$CuO + H_2 \rightarrow Cu + H_2O$			
$3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$			

Q26. Write the balanced equation for the reaction of iron with steam. What colour change is observed at the iron surface? Name the type of reaction. (PYQ 2022)

Q27. Write the balanced equations for the following reactions. Also state one observation in each case: (PYQ 2020)

Reaction	Balanced Equation	Observation
Lead nitrate crystals heated in test tube		
Magnesium ribbon burnt in air		
Zinc added to dilute H_2SO_4		

Q28. What are the various ways in which a chemical reaction can be identified? State four characteristic features with one example each. (PYQ 2019)

Q29. Write balanced equations for: (i) electrolytic decomposition of water (ii) thermal decomposition of calcium carbonate (iii) photolytic decomposition of silver chloride. Name the type of decomposition reaction in each case. (PYQ)

Q30. Why is respiration called an exothermic process? Why is photosynthesis called an endothermic process? Write balanced equations for both. (PYQ 2018)

SECTION C - 5 MARKS

Q1. Write balanced chemical equations for the following reactions with state symbols. Classify each reaction: (PYQ 2019)

Reaction	Balanced Equation (with State Symbols)	Type
Zinc + silver nitrate solution -> zinc nitrate + silver		
Calcium oxide + water -> calcium hydroxide		
Mercuric oxide heated -> mercury + oxygen		
Barium chloride + sodium sulphate -> barium sulphate + sodium chloride		
Ferric oxide + carbon monoxide -> iron + carbon dioxide		

Q2. For each of the following reactions, identify the type, write the balanced chemical equation with state symbols, and state one observation: (PYQ 2020)

Reaction	Balanced Equation	Type	Observation
Iron nail placed in blue copper sulphate solution			
Sodium sulphate + barium chloride solutions mixed			
Silver chloride exposed to sunlight			
Lime water + CO_2			
Lead nitrate crystals heated in test tube			

Q3. Identify the substance oxidised, reduced, oxidising agent, and reducing agent in each of the following: (PYQ 2022)

Reaction	Substance Oxidised	Substance Reduced	Oxidising Agent	Reducing Agent
$CuO + H_2 \rightarrow Cu + H_2O$				
$2PbO + C \rightarrow 2Pb + CO_2$				

Reaction	Substance Oxidised	Substance Reduced	Oxidising Agent	Reducing Agent
$MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$				
$ZnO + C \rightarrow Zn + CO$				
$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$				

Q4. Write the balanced equations for the following decomposition reactions. Classify each as thermal, electrolytic, or photolytic: (PYQ 2019)

Substance Decomposed	Balanced Equation	Type of Decomposition	One Use
Ferrous sulphate			
Water (by electricity)			
Silver bromide (by light)			
Lead nitrate (by heat)			
Calcium carbonate (by heat)			

Q5. Complete the following table by writing the balanced chemical equation, identifying the type, and naming the products: (PYQ 2023)

Reactants	Balanced Equation	Type of Reaction	Products
$Mg + O_2$			
$Zn + CuSO_4(aq)$			
$2KBr + BaI_2$			
$3Fe + 4H_2O$ (steam)			
$Na_2SO_4(aq) + BaCl_2(aq)$			

Q6. Write balanced chemical equations for the following. State the type of each reaction and one observation: (PYQ 2024)

Reaction	Balanced Equation	Type	Observation
Copper + $2AgNO_3$ solution			
$NaOH + HCl$			

Reaction	Balanced Equation	Type	Observation
$CaCO_3$ heated strongly			
$BaCl_2(aq) + H_2SO_4(aq)$			
$2AgCl$ exposed to sunlight			

Q7. A student performs the following experiments. For each, state what is observed, write a balanced chemical equation, and name the type of reaction: (PYQ 2020)

Experiment	Observation	Balanced Equation	Type
Iron nail placed in blue $CuSO_4$ solution for 20 minutes			
Zinc granules added to dilute H_2SO_4			
Magnesium ribbon burnt in air			
Few drops of NaOH added to $FeSO_4$ solution			
Silver nitrate + sodium chloride solutions mixed			

Q8. Define corrosion and rancidity. Explain the chemical reactions involved in each. State two harmful effects of each and describe three methods to prevent each. (PYQ 2019)

Q9. Answer the following questions based on chemical reactions: (PYQ 2022)

Question	Answer
Name the type of reaction when magnesium combines with oxygen	
Name the type of reaction: $CaO + H_2O \rightarrow Ca(OH)_2$ + heat	
Identify the oxidising agent in: $2PbO + C \rightarrow 2Pb + CO_2$	
Why are antioxidants added to packaged food?	
Name the products when lead nitrate is heated	

Q10. For each of the following descriptions, write the balanced chemical equation, identify the type of reaction, and give one real-life example: (PYQ HOTS)

Description	Balanced Equation	Type	Real-Life Example
A metal combines with oxygen to form its oxide			
A compound breaks down into simpler substances by heat			
A more reactive metal displaces a less reactive metal from salt solution			
Two salt solutions react to form an insoluble precipitate			
A reaction that releases heat into the surroundings			

Q11. Write balanced chemical equations for each of the following and state whether each is exothermic or endothermic: (PYQ 2019)

Reaction	Balanced Equation	Type	Exothermic / Endothermic
Combination of hydrogen and oxygen			
Thermal decomposition of calcium carbonate			
Iron reacts with copper sulphate solution			
Barium chloride reacts with sodium sulphate			
Burning of natural gas (methane)			

Q12. Write the balanced chemical equations for the following reactions. Identify the oxidising agent and reducing agent in each: (PYQ 2020)

Reaction	Balanced Equation	Oxidising Agent	Reducing Agent
Hydrogen sulphide + chlorine -> hydrochloric acid + sulphur			
Copper oxide + hydrogen -> copper + water			

Reaction	Balanced Equation	Oxidising Agent	Reducing Agent
Lead oxide + carbon -> lead + carbon monoxide			
Iron (III) oxide + carbon monoxide -> iron + carbon dioxide			
Manganese dioxide + hydrochloric acid			

Q13. State what you observe and write the balanced chemical equation when: (PYQ 2022)

Situation	Observation	Balanced Chemical Equation
Copper powder is heated in a china dish		
A silver article is left in air for several days		
Iron is left in a moist atmosphere for a long time		
Copper wire is dipped into silver nitrate solution		
Ferrous sulphate crystals are heated in a dry test tube		

Q14. Complete and balance the following chemical equations. Write the type of reaction and state one observation for each: (PYQ 2023)

Incomplete Equation	Balanced Equation	Type	Observation
Zn + H2SO4 ->			
CaCO3 -> (heat)			
AgNO3(aq) + NaCl(aq) ->			
Na + H2O ->			
Al + CuCl2 ->			

Q15. A student is given the following chemical reactions to analyse. For each, state the type, write the balanced equation, and answer the sub-question: (PYQ 2024 case-based)

Reaction	Balanced Equation	Type	Sub-question
Pb(NO3)2 heated			Name the brown gas evolved
BaCl2 + H2SO4(aq)			Name the white insoluble product
Mg ribbon burnt in air			Name the type of oxide formed
Fe + dilute HCl			Name the gas evolved
2AgBr -> 2Ag + Br2 (light)			Name one use of this reaction